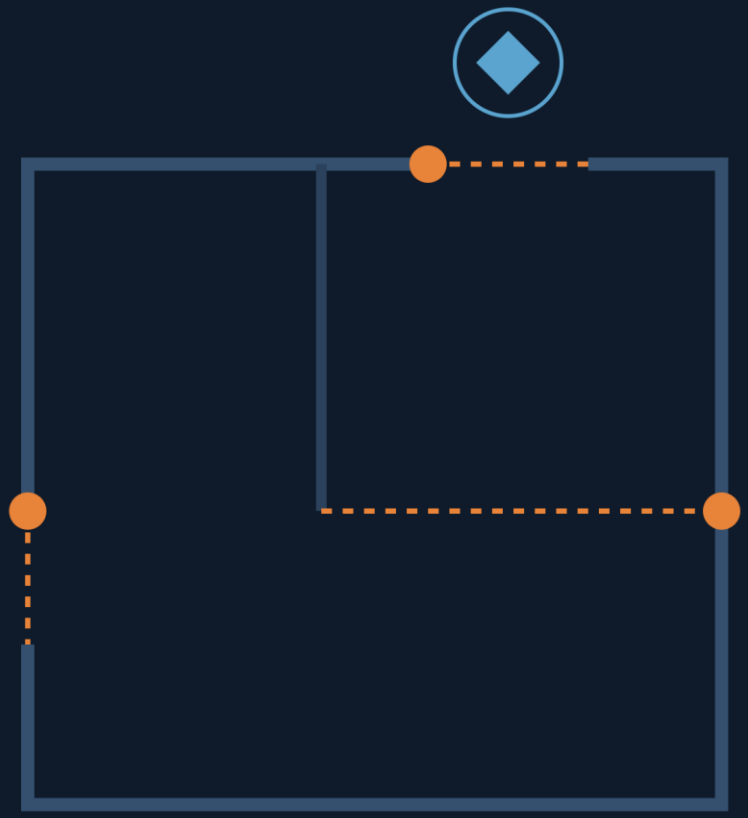
EUDIS AUTUMN 2026 · CHALLENGE 3 · C-UAS

Perimeter Sentry



A sentry node for every window, door and corridor. Ultrasound catches what passes, microphones tell if it’s a drone, and position shows on the helmet map.

Stavros Kedaritis · George Athanasiades · ARCHILOCHUS · 17/10/2026

THE PROBLEM

Small drones are hard to see

Hard to see

Small, low and often behind buildings or trees.

Difficult in operations

You can't carry with you all the Counter-Drone systems

Flexible system

You need a system so that it doesn’t matter if something gets left behind.

OUR SOLUTION

A small, portable system that watches your back





Ultrasound

Catches anything that passes

Microphones

Hear the drone coming

Helmet map

The position appears on the map of the soldier.


mics
Ultra sound

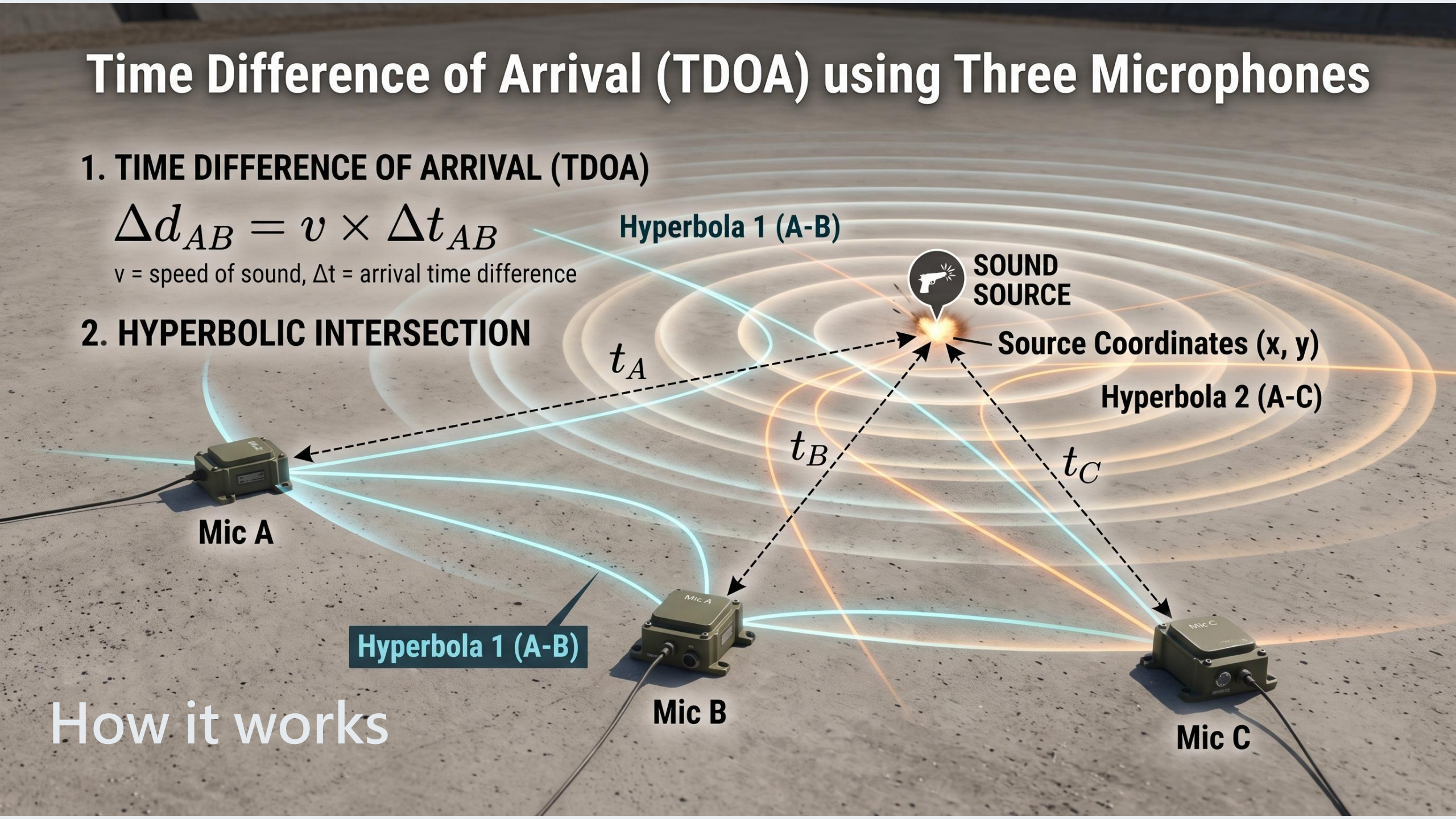
Time Difference of Arrival (TDOA) using Three Microphones

1. TIME DIFFERENCE OF ARRIVAL (TDOA)

$$\Delta d_{AB} = v \times \Delta t_{AB}$$

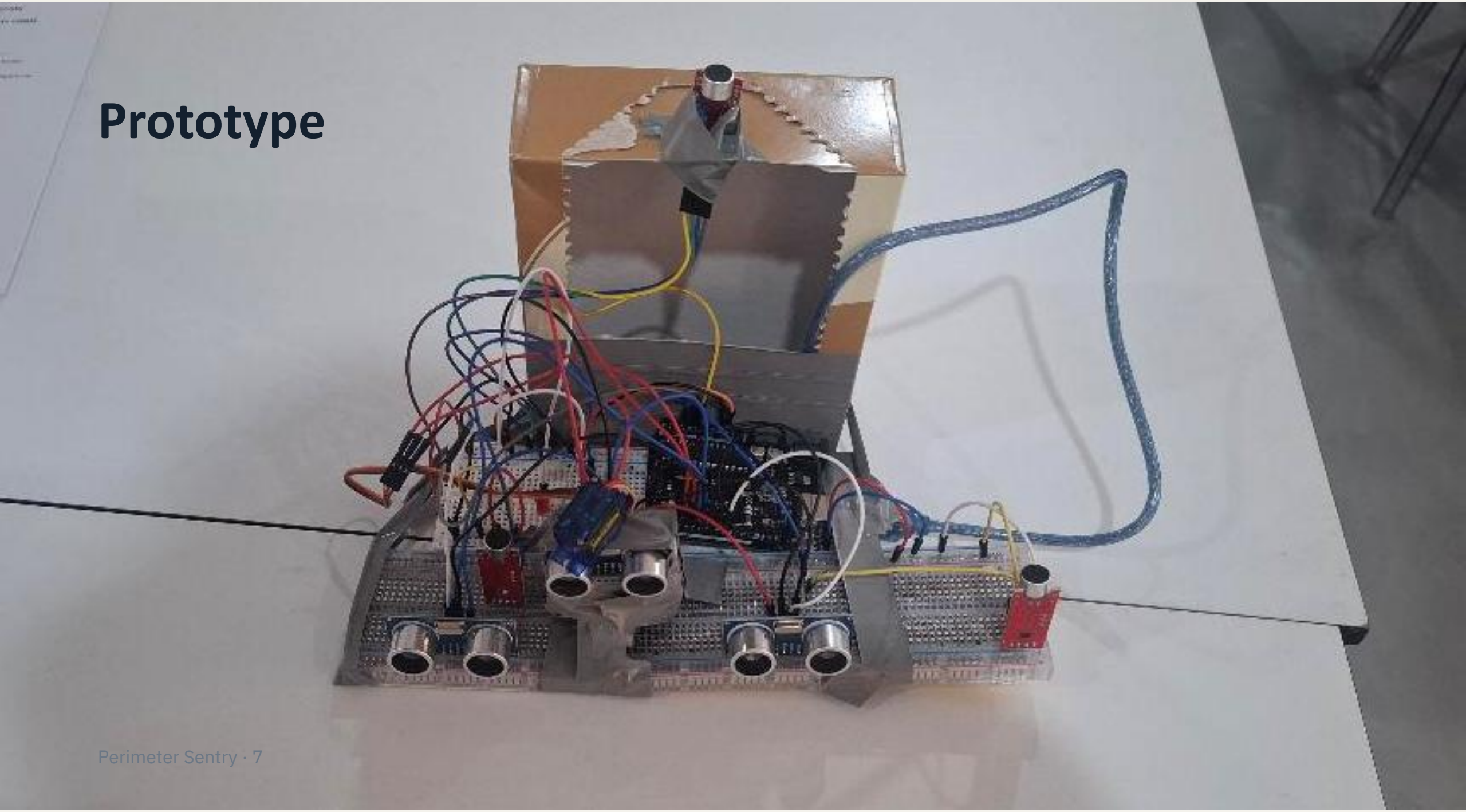
v = speed of sound, Δt = arrival time difference

2. HYPERBOLIC INTERSECTION



How it works

Prototype



What we need:

- **Test access on the battlefield**

Securing immediate entry to the field is essential for gathering accurate, actionable intelligence and maintaining effective coordination across all active units.

PERIMETER SENTRY

Hear it before it arrives.

A sentry on every way in, and the answer on every helmet.

Stavros Kedaritis · George Athanasiades · ARCHILOCHUS · stavrosked@gmail.com